

CSP

Converting problems to CSPs

- A problem to be converted to CSP requires the following steps:
- **Step 1:** Create a variable set.(Representing state)
- **Step 2:** Create a domain set.(values)
- **Step 3:** Create a constraint set with variables and domains after considering the constraints.(limitations)

	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

VARIABLES – variables are cells (A1-I9)

DOMAIN – domain of each variable is {1,2,3,4,5,6,7,8,9}

CONSTRAINT– constraints: rows, columns, boxes contain all different numbers

Varieties of constraints

- **Unary constraints** involve a single variable, e.g. SA ≠ green
- **Binary constraints** involve pairs of variables (relates two variables),e.g., SA ≠ WA
- **Higher-order constraints** involve 3 or more variables, e.g. cryptarithmic puzzles (column constraints)

Backtracking search

- Depth-first search for CSPs with single-variable assignments is called backtracking search
- Variable assignments are commutative: [WA = red then NT = green] same as [NT = green then WA = red]
- Only need to consider assignments to a single variable at each node
- Backtracks when variable has no legal value

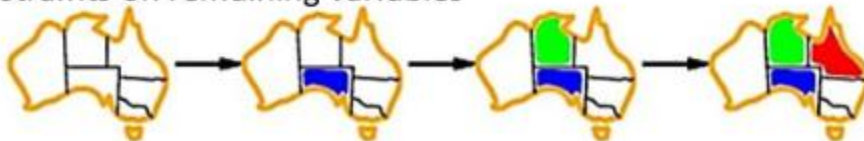
MRV (Minimum Remaining Values)

(MRV)

This intuitive idea-choosing the variable with the fewest "legal" values is called the minimum remaining values (MRV) heuristic (also known as most constrained variable)

Most-Constraining Variable (Degree Heuristics)

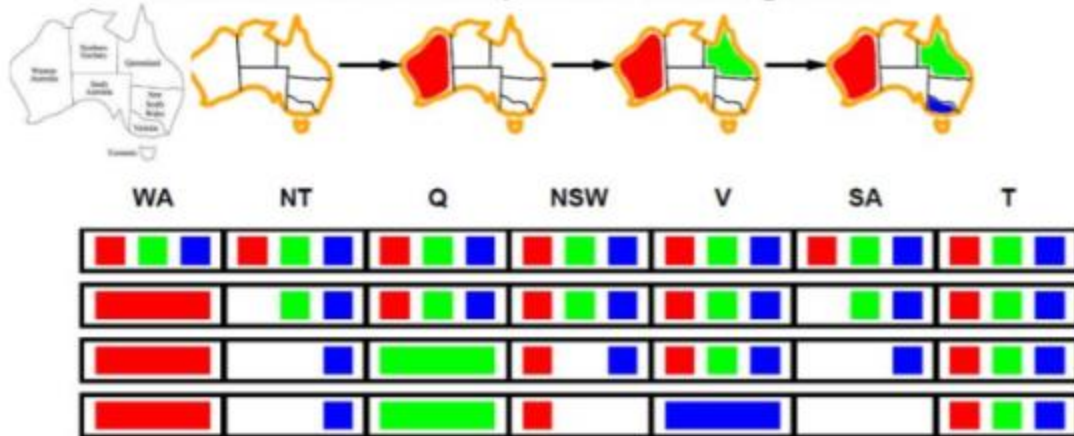
- Most constraining variable: choose the variable with the most constraints on remaining variables



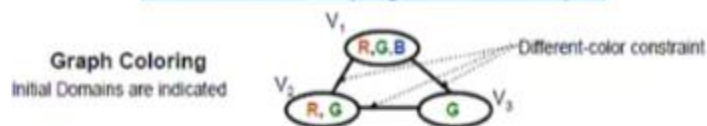
Forward Checking

Idea: Keep track of remaining legal values for unassigned variables

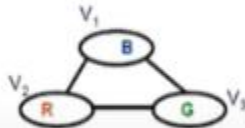
Terminate search when any variable has no legal values



Constraint Propagation Example

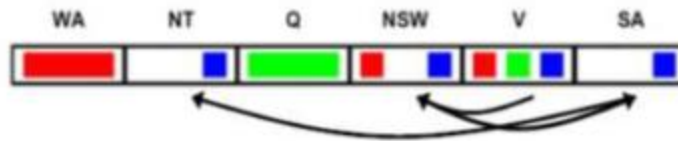


Arc examined	Value deleted
$V_1 - V_2$	none
$V_1 - V_3$	$V_1(G)$
$V_2 - V_3$	$V_2(G)$
$V_1 - V_2$	$V_1(R)$
$V_1 - V_3$	none
$V_2 - V_3$	none



ARC Consistency

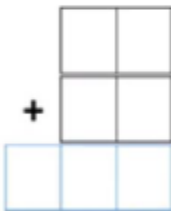
- A simple form of propagation makes sure **all** arcs are simultaneously consistent:



- Arc consistency detects failure earlier than forward checking
- Important: If X loses a value, neighbors of X need to be rechecked!
- Must rerun after each assignment!

*Remember:
Delete from
the tail!*

Cryptarithmic Problem



$$\begin{array}{r}
 \text{T} \text{ O} \\
 + \text{G} \text{ O} \\
 \hline
 \text{O} \text{ U} \text{ T}
 \end{array}$$

Character	Code
T	
O	
G	
U	

ADVERSARIAL SEARCH

Zero-Sum Game

- Zero-sum games are adversarial search which involves pure competition.
- In Zero-sum game each agent's gain or loss of utility is exactly balanced by the losses or gains of utility of another agent.
- One player of the game try to maximize one single value, while other player tries to minimize it.

Properties of Mini-Max algorithm:

- **Complete**- Min-Max algorithm is Complete. It will definitely find a solution (if exist), in the finite search tree.
- **Optimal**- Min-Max algorithm is optimal if both opponents are playing optimally.
- **Time complexity**- As it performs DFS for the game-tree, so the time complexity of Min-Max algorithm is $O(b^d)$, where b is branching factor of the game-tree, and m is the maximum depth of the tree.
- **Space Complexity**- Space complexity of Mini-max algorithm is also similar to DFS which is $O(b^d)$,

Alpha Beta Pruning

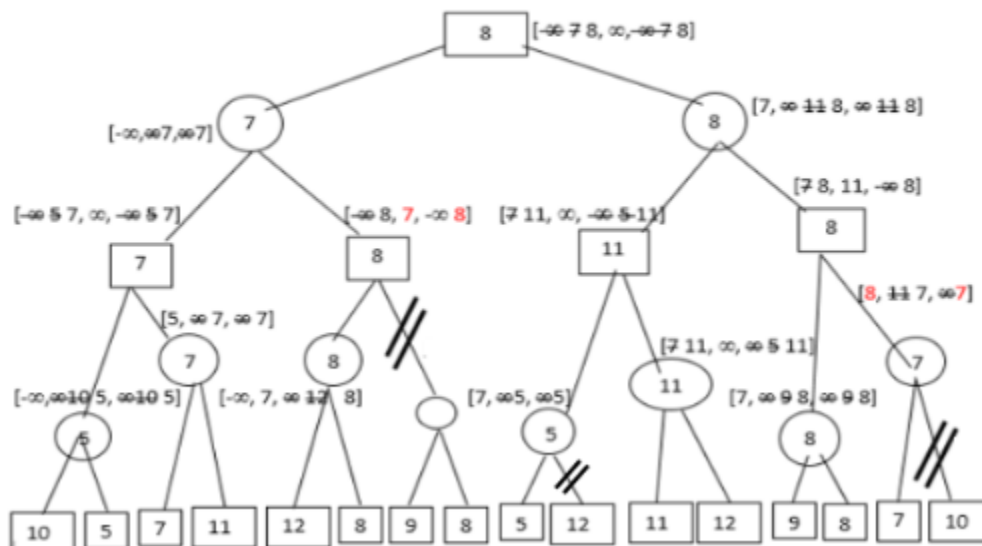
- It is an optimization technique for the min-max algorithm.
- In minmax search, algorithm has to traverse all of the possible game states in depth of the tree i.e. $O(b^d)$ to make decision.
- α - β cuts off branches in the game tree which need not be searched because there already exists a better move available.

The two-parameter can be defined as:

- α = the value of the best (i.e., highest-value) choice we have found so far at any choice point along the path for MAX.
- β = the value of the best (i.e., lowest-value) choice we have found so far at any choice point along the path for MIN.

For MAX Nodes, $V = \max(\alpha)$ & $\alpha \geq -\infty$

For MIN Nodes, $V = \min(\beta)$ & $\beta \geq \infty$



UNERTAINTY

- **Definition of Conditional Probability:**

$$P(a \mid b) = P(a \wedge b) / P(b)$$

- **Product rule gives an alternative formulation:**

$$\begin{aligned} P(a \wedge b) &= P(a \mid b) * P(b) \\ &= P(b \mid a) * P(a) \end{aligned}$$

- **A general version holds for whole distributions:**

$$P(\text{Weather}, \text{Cavity}) = P(\text{Weather} \mid \text{Cavity}) * P(\text{Cavity})$$

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

Probability of any proposition computed by finding atomic events where proposition is true and adding their probabilities

- $P(\text{cavity} \vee \text{toothache})$
 $= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064$
 $= 0.28$
- $P(\text{cavity})$
 $= 0.108 + 0.012 + 0.072 + 0.008$
 $= 0.2$

$P(\text{cavity})$ is called a marginal probability and the process of computing this is called marginalization

$$\begin{aligned} P(\neg \text{cavity} \mid \text{toothache}) &= P(\neg \text{cavity} \wedge \text{toothache}) / P(\text{toothache}) \\ &= (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064) \\ &= 0.4 \end{aligned}$$

Denominator is viewed as a normalization constant:

- Stays constant no matter what the value of Cavity is.
(Book uses α to denote normalization constant $1/P(X)$, for random variable X .)

BAYES RULE

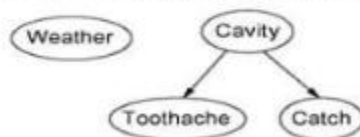
$$P(A | B) = [P(B | A) * P(A)] / P(B)$$

PROBABLISTIC REASONING

Bayesian networks

- A simple, **graphical notation** for conditional independence assertions and hence for **compact** specification of full joint distributions
- A directed, acyclic graph (DAG)
- A set of nodes, one per **variable** (discrete or continuous)
- A set of directed links (arrows) connects pairs of nodes. X is a parent of Y if there is an arrow (**direct influence**) from node X to node Y .
- Each node X_i has a conditional probability distribution $P(X_i | Parents(X_i))$ that quantifies the effect of the parents on the node.
- Combinations of the **topology** and the **conditional distributions** specify (implicitly) the full joint distribution for all the variables.

Topology of network encodes conditional independence assertions:

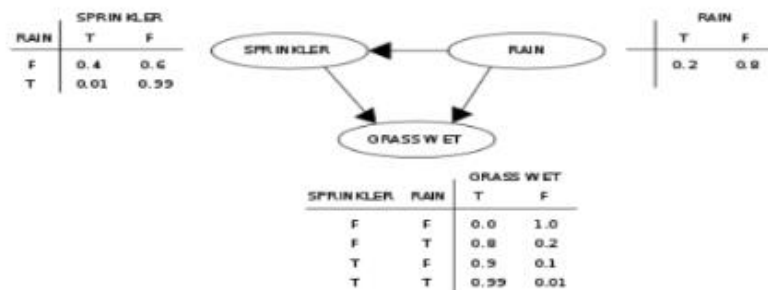


Weather is independent of the other variables

Toothache and *Catch* are conditionally independent given *Cavity*

Example:2

Two events can cause grass to be wet: an active sprinkler or rain. Rain has a direct effect on the use of the sprinkler (namely that when it rains, the sprinkler usually is not active). This situation can be modeled with a Bayesian network. Each variable has two possible values, T (for true) and F (for false).



What is the probability that it is raining, given the grass is wet?

Using the expansion for the joint probability function $\Pr(G, S, R)$ and the conditional probabilities from CPT

$$\begin{aligned}
 \Pr(G = T, S = T, R = T) &= \Pr(G = T | S = T, R = T) \Pr(S = T | R = T) \Pr(R = T) \\
 &= 0.99 \times 0.01 \times 0.2 \\
 &= 0.00198.
 \end{aligned}$$

$$\Pr(R = T | G = T) = \frac{0.00198_{TTT} + 0.1584_{TFT}}{0.00198_{TTT} + 0.288_{TTF} + 0.1584_{TFT} + 0.0_{TFF}} = \frac{891}{2491} \approx 35.77\%.$$

Q1: You, your Father and Mother lineup randomly in a queue to take a memorable picture at your Convocation. Find the $P(A/B)$ such that

A= Daughter on one end, B= Father in Middle

1. Given Condition (Event B):

The Father is in the middle. There are only 2 possible arrangements where this is true:

- (Daughter, **Father**, Mother)
- (Mother, **Father**, Daughter)
- Therefore, $n(B) = 2$.

2. Target Event (Event A given B):

Out of those 2 specific arrangements, in how many is the Daughter on one end?

- In both cases—(D, F, M) and (M, F, D)—the Daughter is at an end.
- Therefore, $n(A \cap B) = 2$.

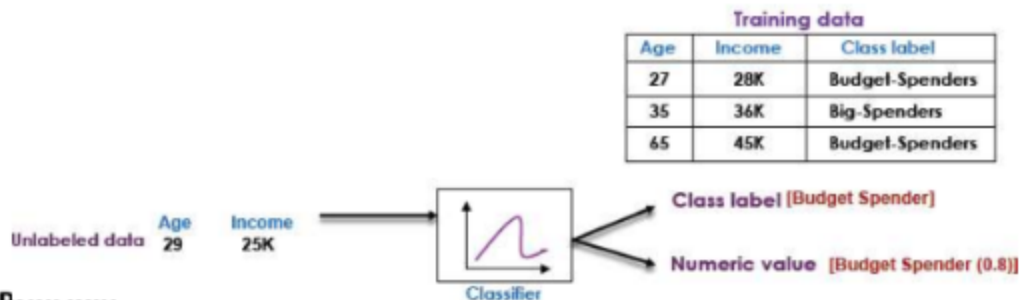
3. Calculation:

Using the conditional probability formula:

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{2}{2} = 1$$

Definition

- Classification is also called **Supervised Learning**
- **Supervision**
 - The training data (observations, measurements, etc) are used to learn a classifier
 - The training data are **labeled data**
 - New data (**unlabeled**) are **classified** Using the training data



- **Principle**
 - Construct models (functions) based on some training examples
 - Describe and distinguish classes or concepts for future prediction
 - Predict some unknown class labels

Classification—A Three-Step Process

- **Model construction**: describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- **Test Classifier** (Estimate accuracy of the model)
 - The known label of **test sample** is compared with the classified result from the model
 - Accuracy rate is the **percentage** of test set samples that are correctly classified by the model
 - Test set is **independent** of training set, otherwise **over-fitting** will occur
- **Model usage**: for classifying future or unknown objects
 - If the accuracy is acceptable, use the model to **classify data tuples** whose class labels are not known

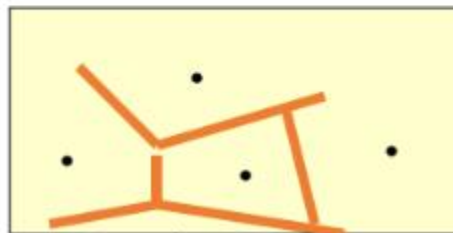
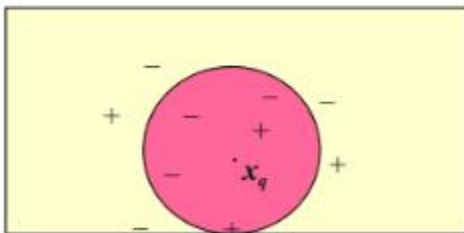
Naïve Bayesian Classifier: An Example

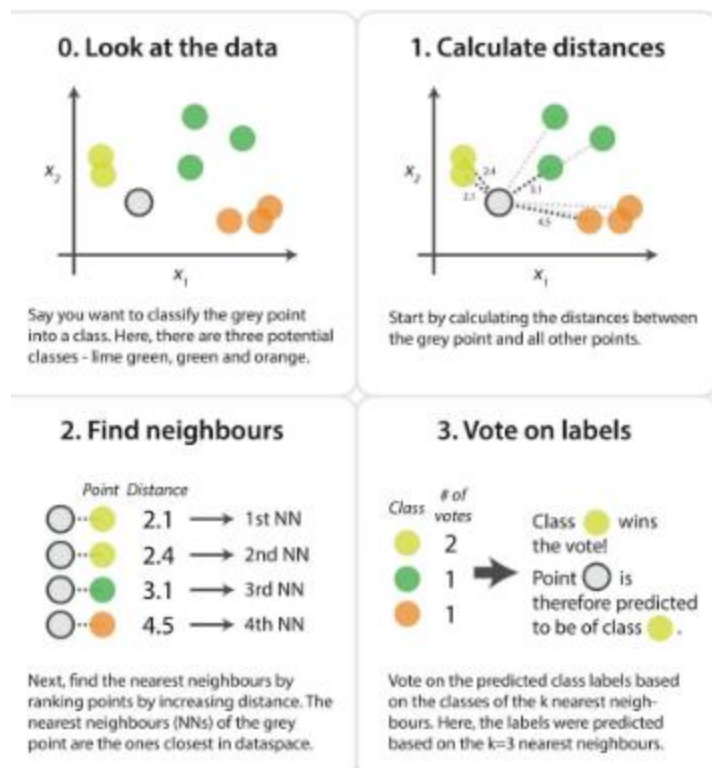
age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$
- Likelihood of "yes" = $2/9 \times 4/9 \times 6/9 \times 9/14$
= 0.02821
- Likelihood of "no" = $3/5 \times 2/5 \times 1/5 \times 2/5 \times 5/14$
= 0.0068
- Turn into probability by normalizing likelihood values
- probability of "yes" = $0.02821 / (0.02821 + 0.0068)$
= 0.8057
- probability of "no" = $0.0068 / (0.02821 + 0.0068)$
= 0.1942

The k -Nearest Neighbor (k -NN) Algorithm

- All instances correspond to points in the n -D space
- The nearest neighbor are defined in terms of Euclidean distance, $\text{dist}(X_1, X_2)$
- Target function could be discrete- or real- valued
- For discrete-valued, k -NN returns the most common value (= majority vote) among the k training examples nearest to x_q
- Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples





Linear regression: involves a response variable y and a single predictor variable x

$$y = w_0 + w_1 x$$

where w_0 (y-intercept) and w_1 (slope) are regression coefficients

Method of least squares: estimates the best-fitting straight line

$$w_1 = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|} (x_i - \bar{x})^2} \quad w_0 = \bar{y} - w_1 \bar{x}$$

Multiple linear regression: involves more than one predictor variable

- Training data is of the form $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_{|D|}, y_{|D|})$
- Ex. For 2-D data, we may have: $y = w_0 + w_1 x_1 + w_2 x_2$
- Solvable by extension of least square method or using SAS, S-Plus
- Many nonlinear functions can be transformed into the above

Example: Linear Regression

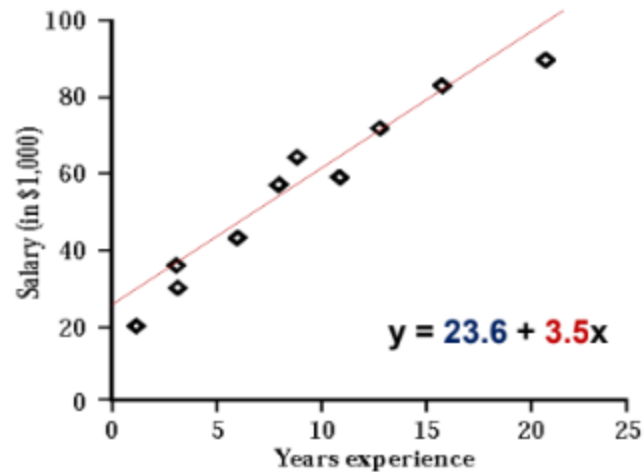
X (Years experience)	Y (Salary)
3	30K
8	57K
9	64K
13	75K
3	36K
6	43K
11	59K
21	90K
1	20K
16	83K

- Given data

$$\bar{x} = 9.1 \quad \bar{y} = 55.4$$

$$w_1 = \frac{(3-9.1)(30-55.4) + \dots + (16-9.1)(83-55.4)}{(3-9.1)^2 + \dots + (16-9.1)^2}$$

$$w_0 = 55.4 - (3.5)(9.1) = 23.6$$



CONFUSION MATRIX

Confusion Matrix

	Predicted		
	NO	Yes	
ACTUAL	165	60	
	NO [TN] ⁵⁰	[FP] ¹⁰	
Yes	[FN] ⁵	[TP] ¹⁰⁰	105
	55	110	

o Accuracy = $\frac{TP + TN}{\text{Total}}$
 $= \frac{(100 + 50)}{165}$
 $= \underline{0.91}$

o Error rate = $1 - \text{accuracy}$
or
 $\frac{FP + FN}{\text{Total}}$
 $= \underline{0.09}$

o Recall :- $\frac{TP}{\text{actual Yes}}$
 $= \frac{100}{105} = \underline{0.95}$

o Precision = $\frac{TP}{\text{Predicted Yes}}$
 $= \frac{100}{110}$
 $= \underline{0.64}$